Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak Dairy C

Physical address of dairy:

25719 Hale RD	Newman	Stanislaus	95360
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X026-X014-X071-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: Landline: ; Cellular: (209) 495-8821

P.O. Box 1331	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Nyman, Brad

Legal owner name: Nyman, Brad

Telephone no.: Landline: ; Cellular: (209) 495-8821

P.O. Box 1331	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	500	0	0	0
Number under roof	600	75	0	0	0	0
Maximum number	600	75	500	0	0	0
Average number	600	75	500	0	0	0
Avg live weight (lbs)	950	1,000	750	0		

Predominant milk cow breed: Jersey

Average milk production: 50 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,750.41 *tons per reporting period*

Total nitrogen from manure: 243,952.04 *lbs per reporting period*

After ammonia losses (30% loss applied): 170,766.43 *lbs per reporting period*

Total phosphorus from manure: 40,517.71 *lbs per reporting period*

Total potassium from manure: 107,021.54 *lbs per reporting period*

Total salt from manure: 299,756.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 49,152,000 *gallons*

Total nitrogen generated: 194,459.06 *lbs*

Total phosphorus generated: 20,925.98 *lbs*

Total potassium generated: 287,546.00 *lbs*

Total salt generated: 1,906,537.42 *lbs*

	49,152,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	49,152,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water
IW #1	Ground water
IW #2	Ground water
IW #3	Ground water
IW #4	Ground water

Source Description	Type
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
River Road 1 & 2	65	65	0	none	X049-X025-X010-XXXX
Stuhr 1 & 4	90	90	1	process wastewater	X026-X014-X049-XXXX
Stuhr 10 & 13	122	122	2	process wastewater	X026-X014-X051-XXXX
Stuhr 11 & 14	101	101	1	process wastewater	X026-X014-X049-XXXX
Stuhr 15	23	23	2	process wastewater	X026-X014-X069-XXXX
Stuhr 2 & 5	98	98	1	process wastewater	X026-X014-X049-XXXX
Stuhr 3 & 6	73	73	1	manure	X026-X014-X049-XXXX
Stuhr 7 & 8	111	111	2	process wastewater	X026-X014-X070-XXXX X026-X014-X071-XXXX
Stuhr 9 & 12	91	91	1	process wastewater	X026-X014-X051-XXXX
Totals for areas that were used for application	709	709	11		
Totals for areas that were not used for application	65	65	0		
Land application area totals	774	774	11		

B. CROPS AND HARVESTS

Stuhr 1 & 4

Field name: Stuhr 1 & 4

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 90 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	565.00 *ton*	As-is		15.6	33,700.00	3,380.00	18,600.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.28	423.12	42.44	233.53	1,229.24

Stuhr 10 & 13

Field name: Stuhr 10 & 13

Stuhr 10 & 13

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 122 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	1,385.00 *ton*	As-is		58.6	7,540.00	1,260.00	13,900.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.35	171.20	28.61	315.60	1,165.58

07/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 122 Plant date: 07/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/08/2023	2,346.00 *ton*	As-is		70.0	4,860.00	1,530.00	8,960.00		5.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	19.23	186.91	58.84	344.59	627.65

Stuhr 11 & 14

Field name: Stuhr 11 & 14

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 101 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	626.00 *ton*	As-is		15.6	33,700.00	3,380.00	18,600.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.20	417.75	41.90	230.57	1,213.62

Stuhr 15

Field name: Stuhr 15

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	400.00 *ton*	As-is		59.6	7,870.00	1,190.00	13,100.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	17.39	273.74	41.39	455.65	1,545.74

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	565.00 *ton*	As-is		69.8	3,650.00	1,280.00	5,390.00		4.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.57	179.33	62.89	264.81	606.85

Stuhr 2 & 5

Field name: Stuhr 2 & 5

11/15/2022: Oats, grain

Crop: Oats, grain Acres planted: 98 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	1,045.00 *ton*	As-is		63.0	7,560.00	1,090.00	11,200.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	10.66	161.23	23.25	238.86	946.90

Stuhr 3 & 6

Field name: Stuhr 3 & 6

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 73 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	840.00 *ton*	As-is		63.6	7,040.00	1,060.00	12,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.51	162.02	24.39	280.77	996.86

Stuhr 7 & 8

Field name: Stuhr 7 & 8

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 111 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	1,940.00 *ton*	As-is		62.4	7,400.00	1,110.00	11,400.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	17.48	258.67	38.80	398.49	1,419.45

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 111 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	2,750.00 *ton*	As-is		67.8	4,320.00	1,310.00	5,630.00		4.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.77	214.05	64.91	278.96	684.47

Stuhr 9 & 12

Field name: Stuhr 9 & 12

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 91 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	566.00 *ton*	As-is		17.4	30,500.00	2,480.00	17,300.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.22	379.41	30.85	215.20	1,078.88

NUTRIENT BUDGET

A. LAND APPLICATIONS

Stuhr 1 & 4 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 1 & 4

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	216.78	16.87	423.47	2,143.80	5,184,000.00 *gal*
Application event totals		216.78	16.87	423.47	2,143.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	9.16	0.00	0.00	638.09	7,776,000.00 *gal*
IW #4	Ground water	3.66	0.00	0.00	410.97	7,776,000.00 *gal*
Application event totals		12.81	0.00	0.00	1,049.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.99	0.00	0.00	626.28	7,632,000.00 *gal*
IW #4	Ground water	3.59	0.00	0.00	403.36	7,632,000.00 *gal*
Application event totals		12.58	0.00	0.00	1,029.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.14	0.00	0.00	567.19	6,912,000.00 *gal*
IW #4	Ground water	3.25	0.00	0.00	365.31	6,912,000.00 *gal*
Application event totals		11.39	0.00	0.00	932.50	

Stuhr 1 & 4 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.14	0.00	0.00	567.19	6,912,000.00 *gal*
IW #2	Ground water	8.14	0.00	0.00	567.19	6,912,000.00 *gal*
Application event totals		16.28	0.00	0.00	1,134.39	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.25	0.00	0.00	365.31	6,912,000.00 *gal*
IW #4	Ground water	2.84	0.00	0.00	319.65	6,048,000.00 *gal*
Application event totals		6.09	0.00	0.00	684.96	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.12	0.00	0.00	496.29	6,048,000.00 *gal*
IW #4	Ground water	2.84	0.00	0.00	319.65	6,048,000.00 *gal*
Application event totals		9.97	0.00	0.00	815.94	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.46	0.00	0.00	519.93	6,336,000.00 *gal*
IW #4	Ground water	2.98	0.00	0.00	334.87	6,336,000.00 *gal*
Application event totals		10.44	0.00	0.00	854.80	

10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.46	0.00	0.00	519.93	6,336,000.00 *gal*
IW #4	Ground water	2.98	0.00	0.00	334.87	6,336,000.00 *gal*
Application event totals		10.44	0.00	0.00	854.80	

Stuhr 10 & 13 - 11/15/2022: Oats, silage-soft dough

Stuhr 10 & 13 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 10 & 13

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.38	3.69	92.56	468.59	1,536,000.00 *gal*
IW #2	Ground water	6.00	0.00	0.00	418.42	6,912,000.00 *gal*
IW #3	Ground water	3.37	0.00	0.00	264.76	6,912,000.00 *gal*
Application event totals		56.76	3.69	92.56	1,151.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.19	4.84	121.49	615.02	2,016,000.00 *gal*
Application event totals		62.19	4.84	121.49	615.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.65	13.34	94.45	989.71	1,536,000.00 *gal*
IW #2	Ground water	6.76	0.00	0.00	470.72	7,776,000.00 *gal*
IW #3	Ground water	3.79	0.00	0.00	297.86	7,776,000.00 *gal*
Application event totals		113.19	13.34	94.45	1,758.29	

Stuhr 10 & 13 - 07/12/2023: Corn, silage

Field name: Stuhr 10 & 13

Crop: Corn, silage

Plant date: 07/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Stuhr 10 & 13 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.48	15.01	106.26	1,113.43	1,728,000.00 *gal*
IW #3	Ground water	8.42	0.00	0.00	661.91	17,280,000.00 *gal*
Application event totals		123.90	15.01	106.26	1,775.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	7.15	0.00	0.00	562.62	14,688,000.00 *gal*
Application event totals		7.15	0.00	0.00	562.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.68	8.47	46.89	446.79	4,032,000.00 *gal*
IW #3	Ground water	6.17	0.00	0.00	485.40	12,672,000.00 *gal*
Application event totals		50.85	8.47	46.89	932.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.68	8.47	46.89	446.79	4,032,000.00 *gal*
IW #3	Ground water	6.03	0.00	0.00	474.37	12,384,000.00 *gal*
Application event totals		50.71	8.47	46.89	921.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	5.75	0.00	0.00	452.30	11,808,000.00 *gal*
Application event totals		5.75	0.00	0.00	452.30	

Stuhr 10 & 13 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	7.29	0.00	0.00	573.65	14,976,000.00 *gal*
Application event totals		7.29	0.00	0.00	573.65	

10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	5.61	0.00	0.00	441.27	11,520,000.00 *gal*
Application event totals		5.61	0.00	0.00	441.27	

Stuhr 11 & 14 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 11 & 14

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	214.64	16.70	419.28	2,122.57	5,760,000.00 *gal*
Application event totals		214.64	16.70	419.28	2,122.57	

04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.16	0.00	0.00	568.60	7,776,000.00 *gal*
IW #4	Ground water	3.26	0.00	0.00	366.21	7,776,000.00 *gal*
Application event totals		11.42	0.00	0.00	934.81	

Stuhr 11 & 14 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.01	0.00	0.00	558.07	7,632,000.00 *gal*
IW #4	Ground water	3.20	0.00	0.00	359.43	7,632,000.00 *gal*
Application event totals		11.21	0.00	0.00	917.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.25	0.00	0.00	505.42	6,912,000.00 *gal*
IW #4	Ground water	2.90	0.00	0.00	325.52	6,912,000.00 *gal*
Application event totals		10.15	0.00	0.00	830.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.25	0.00	0.00	505.42	6,912,000.00 *gal*
IW #2	Ground water	7.25	0.00	0.00	505.42	6,912,000.00 *gal*
Application event totals		14.51	0.00	0.00	1,010.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.90	0.00	0.00	325.52	6,912,000.00 *gal*
IW #4	Ground water	2.53	0.00	0.00	284.83	6,048,000.00 *gal*
Application event totals		5.43	0.00	0.00	610.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	6.35	0.00	0.00	442.24	6,048,000.00 *gal*
IW #4	Ground water	2.53	0.00	0.00	284.83	6,048,000.00 *gal*
Application event totals		8.88	0.00	0.00	727.08	

Stuhr 11 & 14 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	6.65	0.00	0.00	463.30	6,336,000.00 *gal*
IW #4	Ground water	2.65	0.00	0.00	298.40	6,336,000.00 *gal*
Application event totals		9.30	0.00	0.00	761.70	

Stuhr 15 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 15

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.25	7.34	184.12	932.09	576,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	136.66	2,430,000.00 *gal*
Application event totals		94.25	7.34	184.12	1,068.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.84	4.89	122.75	621.39	384,000.00 *gal*
Application event totals		62.84	4.89	122.75	621.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	170.15	22.12	156.57	1,640.55	480,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	182.21	3,240,000.00 *gal*
Application event totals		170.15	22.12	156.57	1,822.77	

Stuhr 15 - 07/01/2023: Corn, silage

Stuhr 15 - 07/01/2023: Corn, silage

Field name: Stuhr 15

Crop: Corn, silage

Plant date: 07/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	170.15	22.12	156.57	1,640.55	480,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	167.03	2,970,000.00 *gal*
Application event totals		170.15	22.12	156.57	1,807.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	151.84	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	151.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.86	6.42	35.53	338.56	576,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	151.84	2,700,000.00 *gal*
Application event totals		33.86	6.42	35.53	490.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	159.43	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	159.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.86	6.42	35.53	338.56	576,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	136.66	2,430,000.00 *gal*
Application event totals		33.86	6.42	35.53	475.22	

Stuhr 15 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	136.66	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	136.66	

10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	121.47	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	121.47	

Stuhr 2 & 5 - 11/15/2022: Oats, grain

Field name: Stuhr 2 & 5

Crop: Oats, grain

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.24	3.44	86.42	437.51	1,152,000.00 *gal*
IW #2	Ground water	6.54	0.00	0.00	455.78	6,048,000.00 *gal*
IW #4	Ground water	2.61	0.00	0.00	293.55	6,048,000.00 *gal*
Application event totals		53.39	3.44	86.42	1,186.84	

01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.62	4.02	100.83	510.43	1,344,000.00 *gal*
Application event totals		51.62	4.02	100.83	510.43	

Stuhr 2 & 5 - 11/15/2022: Oats, grain

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.84	12.46	88.19	924.07	1,152,000.00 *gal*
IW #2	Ground water	7.47	0.00	0.00	520.89	6,912,000.00 *gal*
IW #4	Ground water	2.98	0.00	0.00	335.49	6,912,000.00 *gal*
Application event totals		106.30	12.46	88.19	1,780.45	

Stuhr 3 & 6 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 3 & 6

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	206.14	46.54	168.20	0.00	800.00 *ton*
Application event totals		206.14	46.54	168.20	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	4.22	0.00	0.00	331.86	5,184,000.00 *gal*
IW #4	Ground water	3.00	0.00	0.00	337.79	5,184,000.00 *gal*
Application event totals		7.22	0.00	0.00	669.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	4.22	0.00	0.00	331.86	5,184,000.00 *gal*
IW #4	Ground water	3.00	0.00	0.00	337.79	5,184,000.00 *gal*
Application event totals		7.22	0.00	0.00	669.65	

Stuhr 7 & 8 - 11/15/2022: Oats, silage-soft dough

Stuhr 7 & 8 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 7 & 8

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.12	6.08	152.60	772.54	2,304,000.00 *gal*
IW #1	Ground water	16.08	0.00	0.00	467.68	6,480,000.00 *gal*
Application event totals		94.20	6.08	152.60	1,240.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.57	3.55	89.02	450.65	1,344,000.00 *gal*
Application event totals		45.57	3.55	89.02	450.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	28.94	0.00	0.00	841.83	11,664,000.00 *gal*
Application event totals		28.94	0.00	0.00	841.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	169.23	22.00	155.72	1,631.69	2,304,000.00 *gal*
Application event totals		169.23	22.00	155.72	1,631.69	

Stuhr 7 & 8 - 07/01/2023: Corn, silage

Field name: Stuhr 7 & 8

Crop: Corn, silage

Plant date: 07/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Stuhr 7 & 8 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.77	13.75	97.33	1,019.80	1,440,000.00 *gal*
IW #1	Ground water	28.94	0.00	0.00	841.83	11,664,000.00 *gal*
Application event totals		134.71	13.75	97.33	1,861.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	23.58	0.00	0.00	685.93	9,504,000.00 *gal*
Application event totals		23.58	0.00	0.00	685.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.05	3.99	22.08	210.46	1,728,000.00 *gal*
IW #1	Ground water	22.51	0.00	0.00	654.75	9,072,000.00 *gal*
Application event totals		43.55	3.99	22.08	865.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	21.97	0.00	0.00	639.16	8,856,000.00 *gal*
Application event totals		21.97	0.00	0.00	639.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.05	3.99	22.08	210.46	1,728,000.00 *gal*
IW #1	Ground water	21.17	0.00	0.00	615.78	8,532,000.00 *gal*
Application event totals		42.21	3.99	22.08	826.24	

Stuhr 7 & 8 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	22.51	0.00	0.00	654.75	9,072,000.00 *gal*
Application event totals		22.51	0.00	0.00	654.75	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	19.29	0.00	0.00	561.22	7,776,000.00 *gal*
Application event totals		19.29	0.00	0.00	561.22	

Stuhr 9 & 12 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 9 & 12

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	238.22	18.54	465.35	2,355.82	5,760,000.00 *gal*
Application event totals		238.22	18.54	465.35	2,355.82	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.71	0.00	0.00	537.59	6,624,000.00 *gal*
IW #4	Ground water	3.08	0.00	0.00	346.24	6,624,000.00 *gal*
Application event totals		10.79	0.00	0.00	883.83	

Stuhr 9 & 12 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.89	0.00	0.00	619.39	7,632,000.00 *gal*
IW #4	Ground water	3.55	0.00	0.00	398.93	7,632,000.00 *gal*
Application event totals		12.44	0.00	0.00	1,018.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.05	0.00	0.00	560.96	6,912,000.00 *gal*
IW #4	Ground water	3.21	0.00	0.00	361.30	6,912,000.00 *gal*
Application event totals		11.26	0.00	0.00	922.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	8.05	0.00	0.00	560.96	6,912,000.00 *gal*
IW #2	Ground water	7.22	0.00	0.00	503.01	6,198,000.00 *gal*
Application event totals		15.27	0.00	0.00	1,063.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.88	0.00	0.00	323.66	6,192,000.00 *gal*
IW #4	Ground water	2.81	0.00	0.00	316.13	6,048,000.00 *gal*
Application event totals		5.69	0.00	0.00	639.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	7.04	0.00	0.00	490.84	6,048,000.00 *gal*
IW #4	Ground water	2.81	0.00	0.00	316.13	6,048,000.00 *gal*
Application event totals		9.86	0.00	0.00	806.97	

Stuhr 9 & 12 - 01/01/2023: Alfalfa, hay

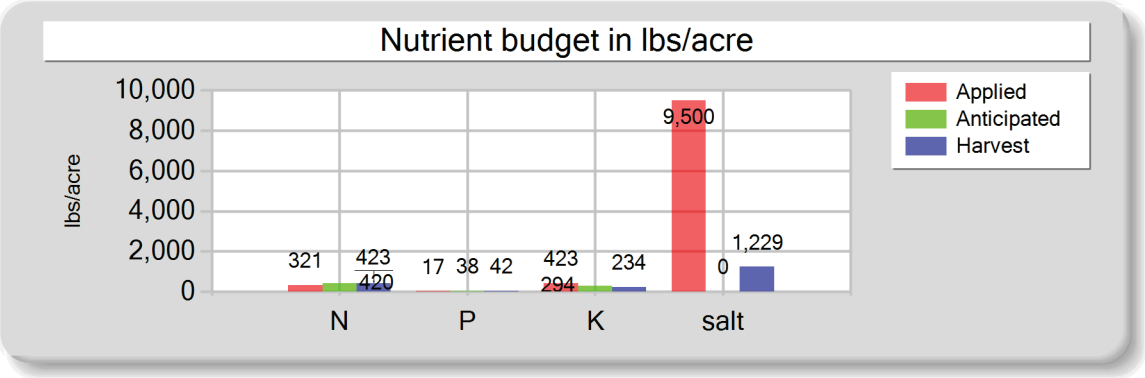
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	6.04	0.00	0.00	420.72	5,184,000.00 *gal*
IW #4	Ground water	2.41	0.00	0.00	270.97	5,184,000.00 *gal*
Application event totals		8.45	0.00	0.00	691.69	

B. NUTRIENT BUDGET

Stuhr 1 & 4 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 1 & 4 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.78	16.87	423.47	2,143.80
Fresh water	89.99	0.00	0.00	7,356.08
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	320.77	16.87	423.47	9,499.88
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	423.12	42.44	233.53	1,229.24
Nutrient balance	-102.35	-25.57	189.94	8,270.64
Applied to removed ratio	0.76	0.40	1.81	7.73

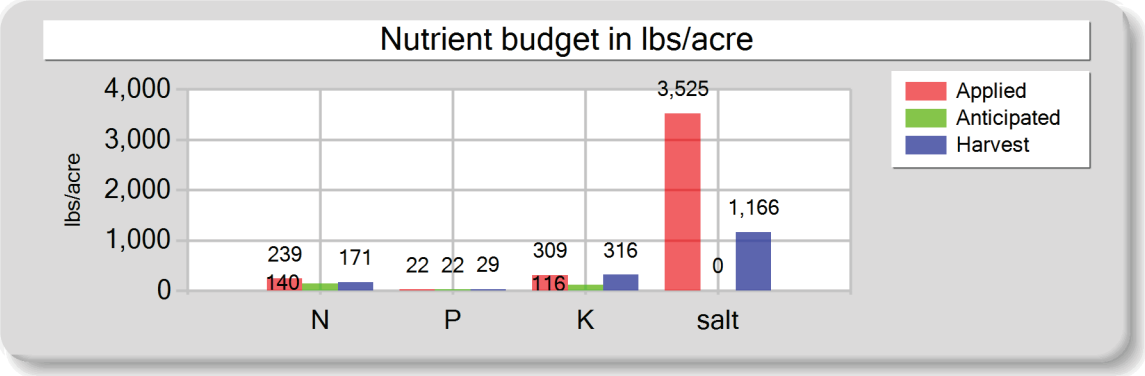
Fresh water applied	
108,864,000.00	*gallons*
4,009.09	*acre-inches*
44.55	*inches/acre*

Process wastewater applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 10 & 13 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 10 & 13 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.22	21.87	308.50	2,073.32
Fresh water	19.91	0.00	0.00	1,451.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.14	21.87	308.50	3,525.09
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	171.20	28.61	315.60	1,165.58
Nutrient balance	67.94	-6.74	-7.09	2,359.51
Applied to removed ratio	1.40	0.76	0.98	3.02

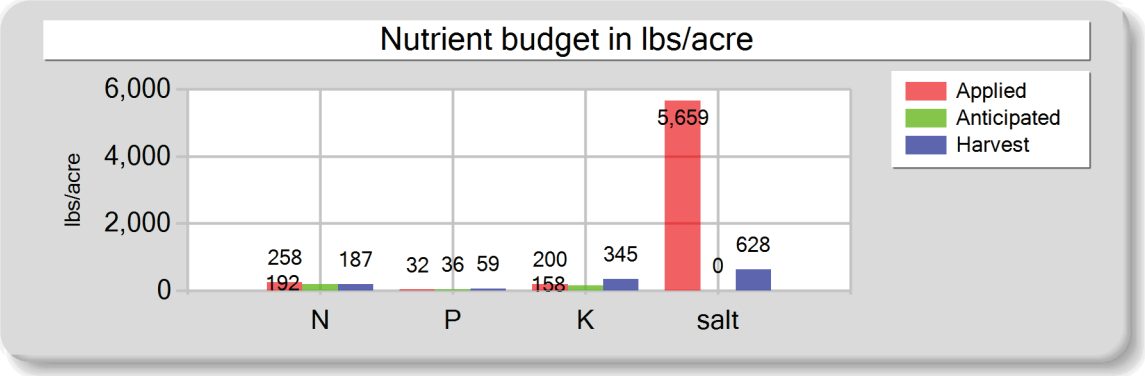
Fresh water applied	
29,376,000.00	*gallons*
1,081.82	*acre-inches*
8.87	*inches/acre*

Process wastewater applied	
5,088,000.00	*gallons*
187.37	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 10 & 13 - 07/12/2023: Corn, silage

Field name: Stuhr 10 & 13 Crop: Corn, silage Plant date: 07/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.84	31.94	200.03	2,007.00
Fresh water	46.43	0.00	0.00	3,651.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.26	31.94	200.03	5,658.53
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	186.91	58.84	344.59	627.65
Nutrient balance	71.35	-26.90	-144.56	5,030.88
Applied to removed ratio	1.38	0.54	0.58	9.02

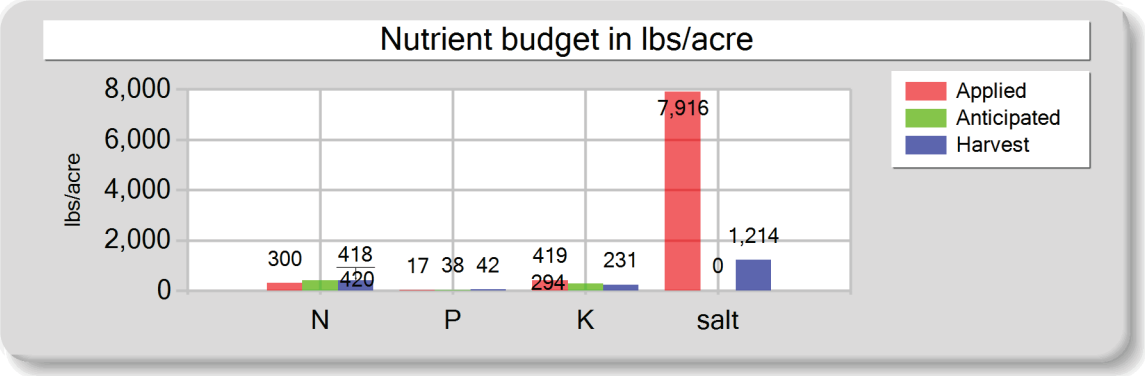
Fresh water applied	
95,328,000.00	*gallons*
3,510.61	*acre-inches*
28.78	*inches/acre*

Process wastewater applied	
9,792,000.00	*gallons*
360.61	*acre-inches*
2.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 11 & 14 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 11 & 14 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.64	16.70	419.28	2,122.57
Fresh water	70.89	0.00	0.00	5,793.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	299.52	16.70	419.28	7,915.80
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	417.75	41.90	230.57	1,213.62
Nutrient balance	-118.22	-25.19	188.71	6,702.18
Applied to removed ratio	0.72	0.40	1.82	6.52

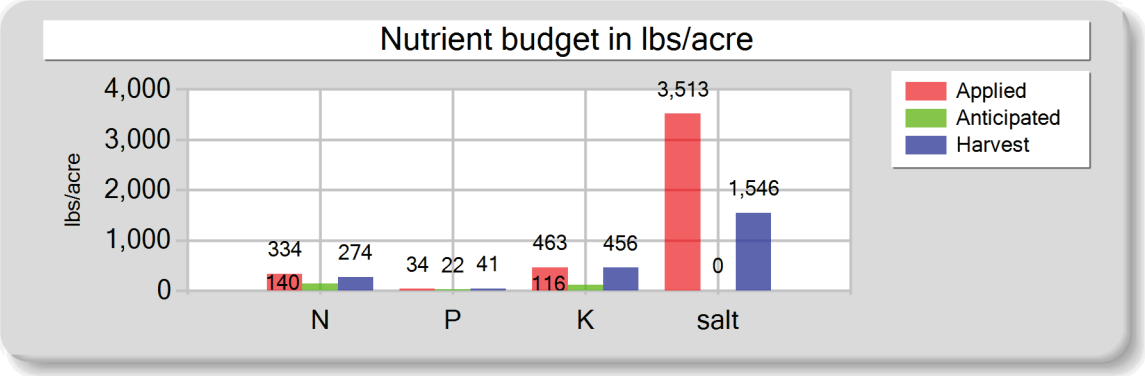
Fresh water applied	
96,192,000.00	*gallons*
3,542.42	*acre-inches*
35.07	*inches/acre*

Process wastewater applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
2.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 15 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 15 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	327.24	34.34	463.43	3,194.03
Fresh water	0.00	0.00	0.00	318.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	334.24	34.34	463.43	3,512.90
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	273.74	41.39	455.65	1,545.74
Nutrient balance	60.50	-7.05	7.78	1,967.16
Applied to removed ratio	1.22	0.83	1.02	2.27

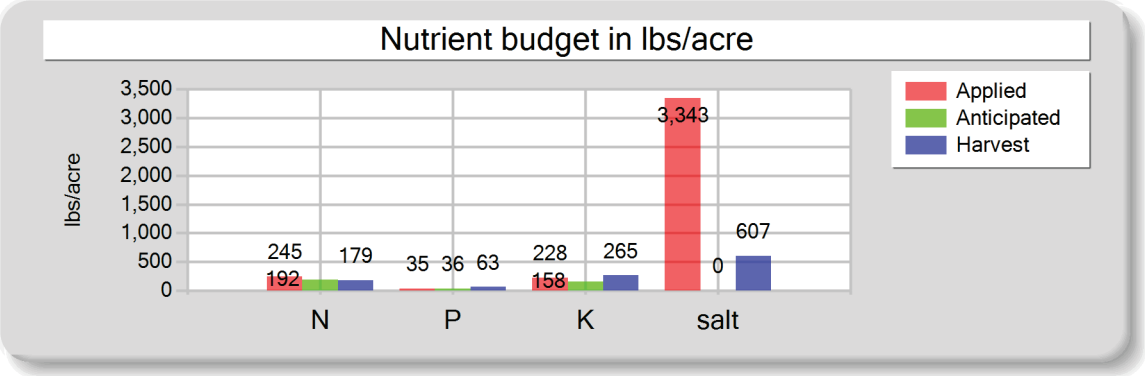
Fresh water applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
9.08	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 15 - 07/01/2023: Corn, silage

Field name: Stuhr 15 Crop: Corn, silage Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.86	34.95	227.62	2,317.67
Fresh water	0.00	0.00	0.00	1,024.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.86	34.95	227.62	3,342.61
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	179.33	62.89	264.81	606.85
Nutrient balance	65.54	-27.94	-37.19	2,735.76
Applied to removed ratio	1.37	0.56	0.86	5.51

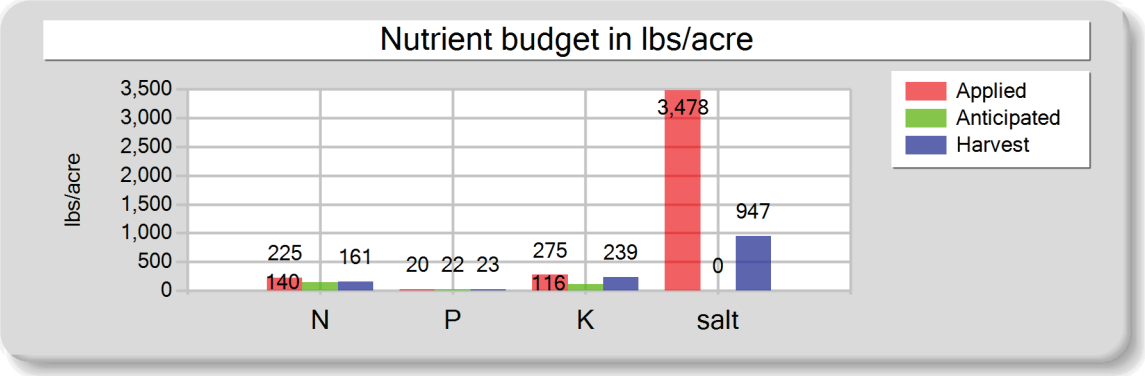
Fresh water applied	
18,225,000.00	*gallons*
671.16	*acre-inches*
29.18	*inches/acre*

Process wastewater applied	
1,632,000.00	*gallons*
60.10	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 2 & 5 - 11/15/2022: Oats, grain

Field name: Stuhr 2 & 5 Crop: Oats, grain Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.70	19.92	275.44	1,872.00
Fresh water	19.61	0.00	0.00	1,605.71
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	225.31	19.92	275.44	3,477.72
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	161.23	23.25	238.86	946.90
Nutrient balance	64.08	-3.33	36.58	2,530.82
Applied to removed ratio	1.40	0.86	1.15	3.67

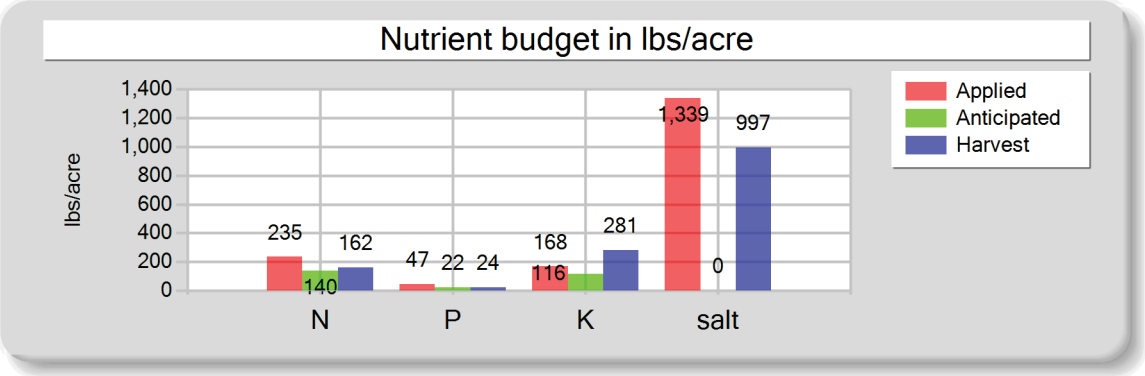
Fresh water applied	
25,920,000.00	*gallons*
954.55	*acre-inches*
9.74	*inches/acre*

Process wastewater applied	
3,648,000.00	*gallons*
134.34	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 3 & 6 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 3 & 6 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	206.14	46.54	168.20	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	14.45	0.00	0.00	1,339.30
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	234.59	46.54	168.20	1,339.30
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	162.02	24.39	280.77	996.86
Nutrient balance	72.57	22.14	-112.57	342.44
Applied to removed ratio	1.45	1.91	0.60	1.34

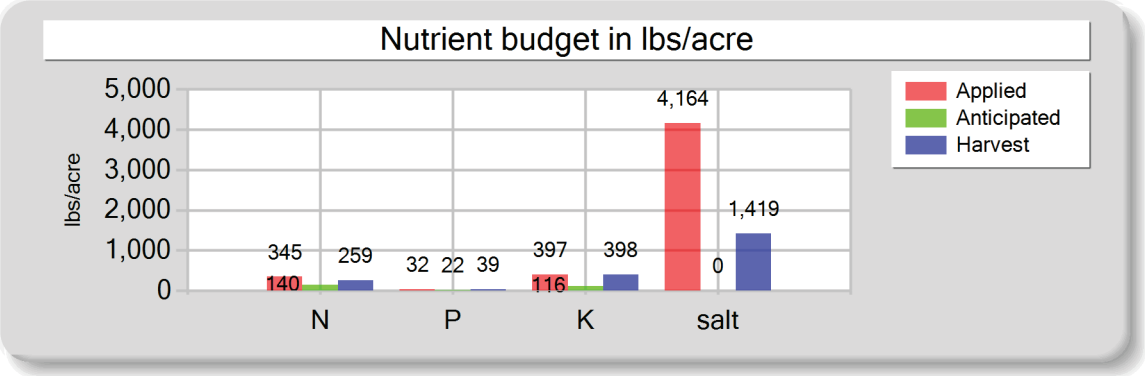
Fresh water applied	
20,736,000.00	*gallons*
763.64	*acre-inches*
10.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 7 & 8 - 11/15/2022: Oats, silage-soft dough

Field name: Stuhr 7 & 8 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.92	31.62	397.34	2,854.87
Fresh water	45.01	0.00	0.00	1,309.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.94	31.62	397.34	4,164.38
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	258.67	38.80	398.49	1,419.45
Nutrient balance	86.27	-7.18	-1.15	2,744.93
Applied to removed ratio	1.33	0.82	1.00	2.93

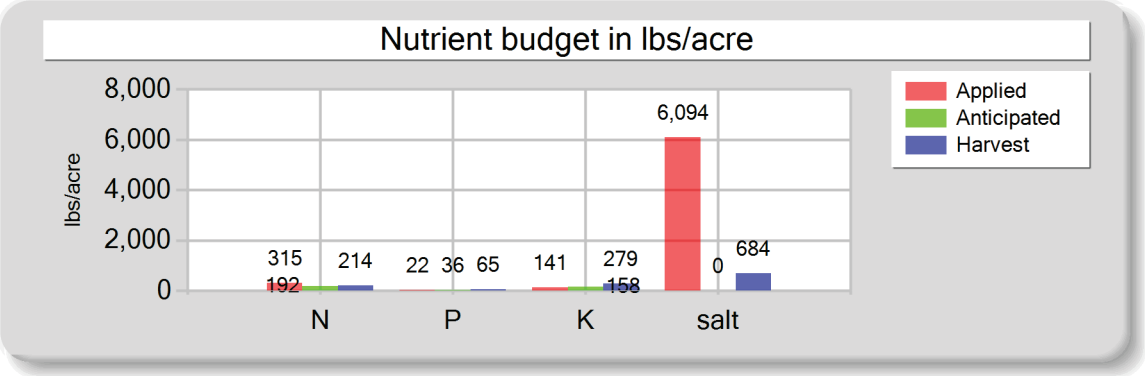
Fresh water applied	
18,144,000.00	*gallons*
668.18	*acre-inches*
6.02	*inches/acre*

Process wastewater applied	
5,952,000.00	*gallons*
219.19	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 7 & 8 - 07/01/2023: Corn, silage

Field name: Stuhr 7 & 8 Crop: Corn, silage Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	147.86	21.73	141.50	1,440.72
Fresh water	159.96	0.00	0.00	4,653.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.82	21.73	141.50	6,094.14
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	214.05	64.91	278.96	684.47
Nutrient balance	100.77	-43.18	-137.47	5,409.67
Applied to removed ratio	1.47	0.33	0.51	8.90

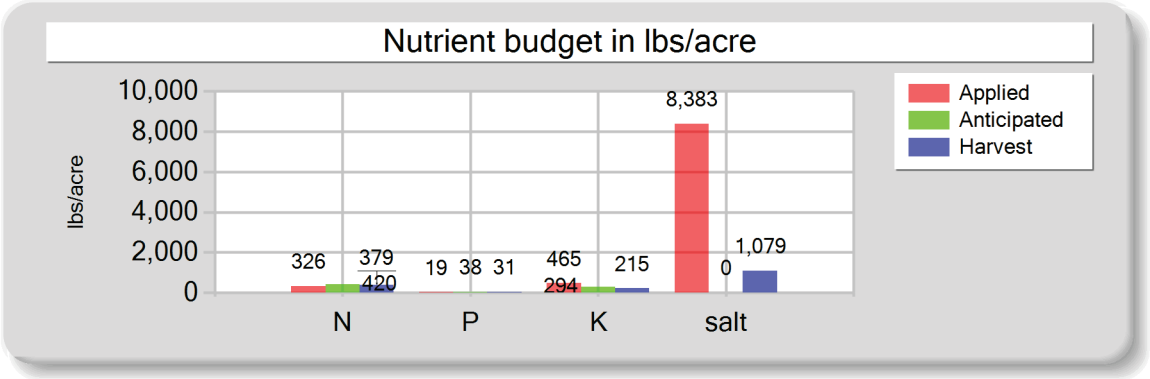
Fresh water applied	
64,476,000.00	*gallons*
2,374.43	*acre-inches*
21.39	*inches/acre*

Process wastewater applied	
4,896,000.00	*gallons*
180.30	*acre-inches*
1.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stuhr 9 & 12 - 01/01/2023: Alfalfa, hay

Field name: Stuhr 9 & 12 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.22	18.54	465.35	2,355.82
Fresh water	73.76	0.00	0.00	6,026.85
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	325.98	18.54	465.35	8,382.67
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	379.41	30.85	215.20	1,078.88
Nutrient balance	-53.43	-12.31	250.15	7,303.78
Applied to removed ratio	0.86	0.60	2.16	7.77

Fresh water applied	
90,150,000.00	*gallons*
3,319.92	*acre-inches*
36.48	*inches/acre*

Process wastewater applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall SM

Sample and source description: Fall SM

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,680.00	13,300.00							0.00
DL	100.00	100.00	100.00							0.01

Spring SM

Sample and source description: Spring SM

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,730.00	13,100.00	9,450.00	8,180.00	2,520.00	2,900.00	684.00		50.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	2,510.00	5,280.00							48.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ3

Sample and source description: WWQ3

Sample date: 09/08/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	451.00	304.00	0.00	0.00	35.10	881.00	175.00	212.00	332.00	3,620.00	0.00	176.00	453.00	7,950.00	4,460
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	100.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,300.00	643.00	0.00	0.00	319.00	12,100.00								9,760.00	6,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	977.00	563.00	0.00	0.00	127.00	899.00								8,340.00	9,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,000.00	714.00	0.00	0.00	61.20	1,000.00								8,080.00	6,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	162.00	47.60	0.00	0.00	30.70	170.00								2,700.00	1,620
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

CCID

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.10								10.00	10

IW #1

IW #1

Sample description: IW #1

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.00	0.00	30.20								1,680.00	960
DL	0.50	0.50	0.50								1.00	10

IW #2

IW #2

IW #2

Sample description: IW #2

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.70	0.00	12.70								1,730.00	885
DL	0.50	0.50	0.50								1.00	10

IW #3

IW #3

Sample description: IW #3

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.12	0.00	7.12								1,100.00	560
DL	0.50	0.50	0.50								1.00	10

IW #4

IW #4

Sample description: IW #4

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.07	0.00	5.07								1,090.00	570
DL	0.50	0.50	0.50								1.00	10

River

River

IW River

Sample description: IW River

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.87	0.00	5.87								2,430.00	1,460
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

River Road 1 & 2

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			20.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			9.00				
DL			4.00				

River Road 1 & 2

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			10.00				
DL			4.00				

Stuhr 1 & 4

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			16.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

Stuhr 10 & 13

Stuhr 10 & 13

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			13.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			12.00				
DL			4.00				

Stuhr 11 & 14

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			4.00				

Stuhr 11 & 14

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

Stuhr 15

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			77.00				
DL			4.00				

Stuhr 2 & 5

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			17.00				
DL			4.00				

Stuhr 3 & 6

Stuhr 3 & 6

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			24.00				
DL			4.00				

Stuhr 7 & 8

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			4.00				

Stuhr 7 & 8

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			74.00				
DL			4.00				

Stuhr 9 & 12

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			16.00				
DL			4.00				

Soil

Sample and source description: Soil

Sample date: 11/16/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			19.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Stuhr 10 & 13 - 11/15/2022: Oats, silage-soft dough

Stuhr 10 & 13 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,540.00	1,260.00	13,900.00		12.40
DL	100.00	100.00	100.00		0.01

Stuhr 10 & 13 - 07/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,860.00	1,530.00	8,960.00		5.44
DL	100.00	100.00	100.00		0.01

Stuhr 11 & 14 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 15.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,700.00	3,380.00	18,600.00		11.60
DL	100.00	100.00	100.00		0.01

Stuhr 15 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,870.00	1,190.00	13,100.00		11.00
DL	100.00	100.00	100.00		0.01

Stuhr 15 - 07/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,650.00	1,280.00	5,390.00		4.09
DL	100.00	100.00	100.00		0.01

Stuhr 2 & 5 - 11/15/2022: Oats, grain

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,560.00	1,090.00	11,200.00		12.00
DL	100.00	100.00	100.00		0.01

Stuhr 2 & 5 - 11/15/2022: Oats, grain

Forage

Sample and source description: Forage

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	1,090.00	4,520.00		4.20
DL	100.00	100.00	100.00		0.01

Stuhr 3 & 6 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,040.00	1,060.00	12,200.00		11.90
DL	100.00	100.00	100.00		0.01

Stuhr 7 & 8 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,400.00	1,110.00	11,400.00		10.80
DL	10.00	100.00	100.00		0.01

Stuhr 7 & 8 - 07/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	1,310.00	5,630.00		4.29
DL	100.00	100.00	100.00		0.01

Stuhr 9 & 12 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,500.00	2,480.00	17,300.00		10.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

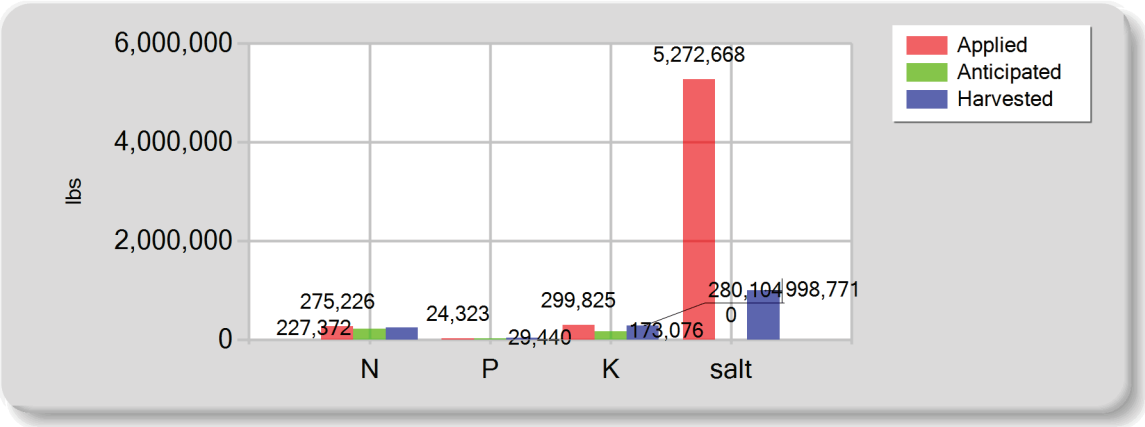
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

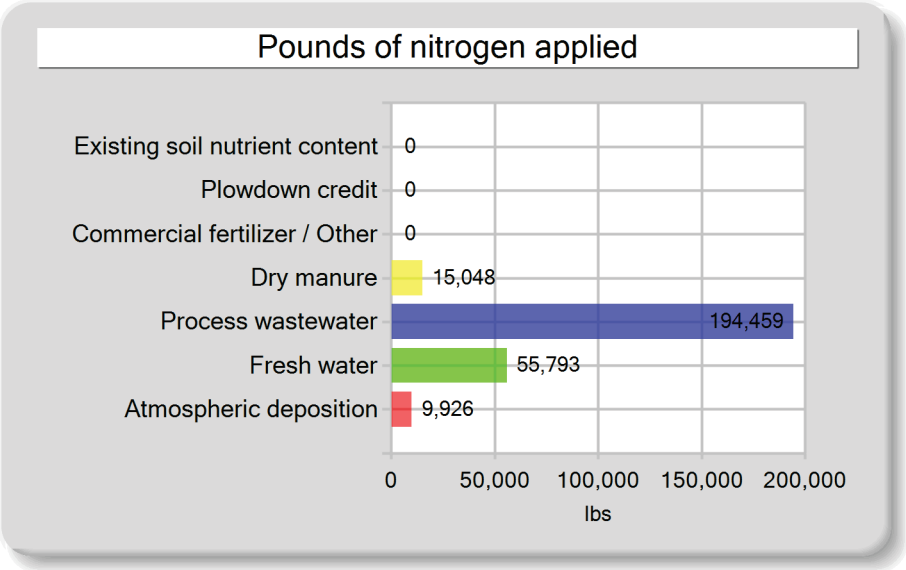
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

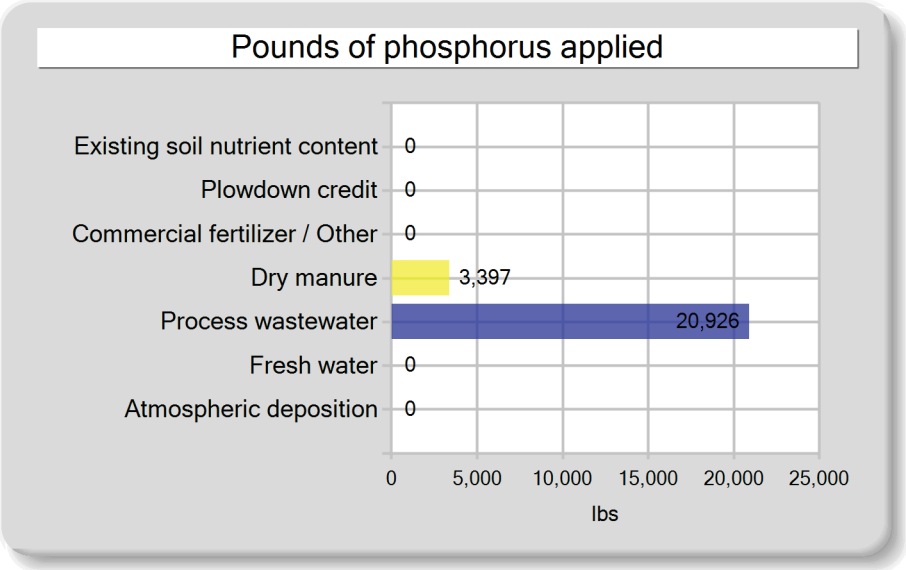
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,048.16	3,397.38	12,278.56	0.00
Process wastewater	194,459.06	20,925.98	287,546.00	1,906,537.42
Fresh water	55,793.11	0.00	0.00	3,366,130.62
Atmospheric deposition	9,926.00	0.00	0.00	0.00
Total nutrients applied	275,226.33	24,323.35	299,824.56	5,272,668.04
Anticipated crop nutrient removal	227,372.00	29,440.40	173,075.80	0.00
Actual crop nutrient removal	249,008.42	39,496.58	280,103.82	998,771.20
Nutrient balance	26,217.91	-15,173.23	19,720.74	4,273,896.84
Applied to removed ratio	1.11	0.62	1.07	5.28

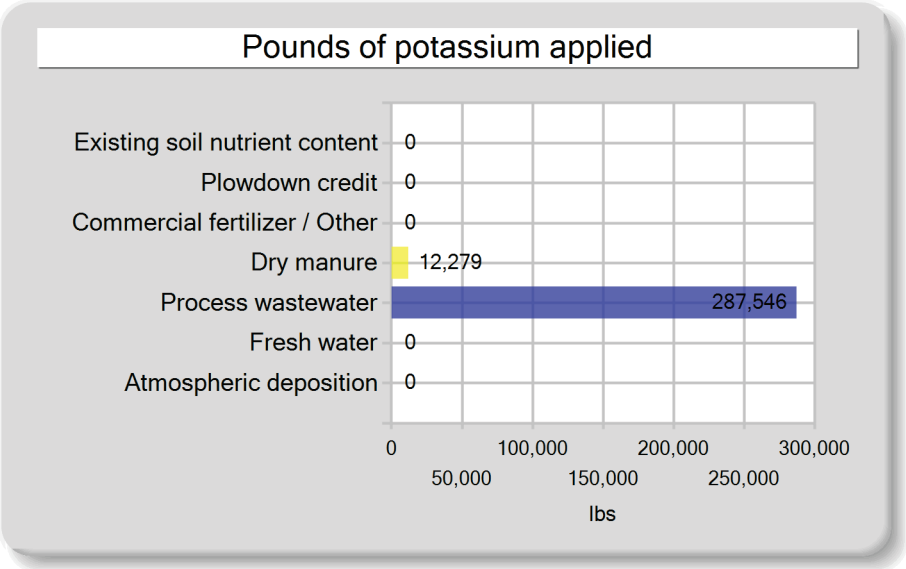
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

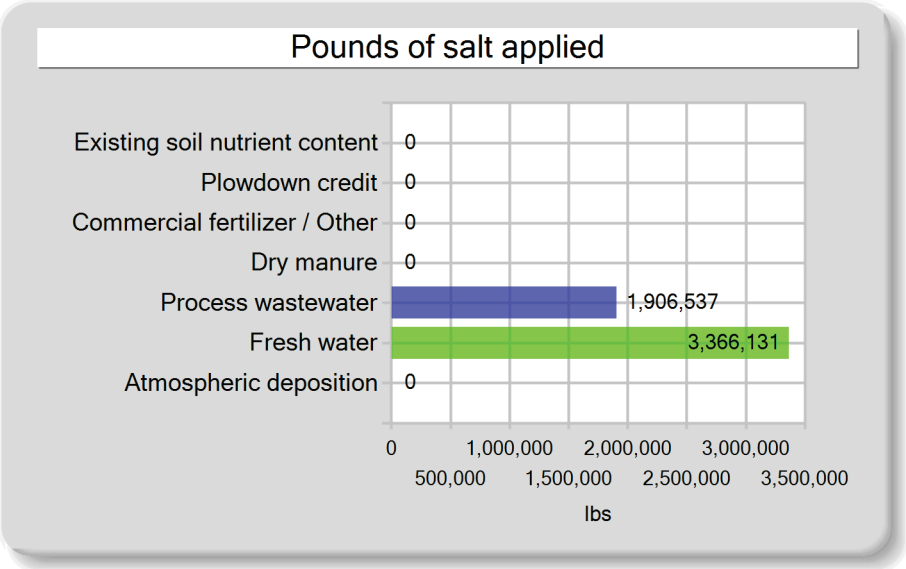


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:28 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nyman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy C
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3240022A	Electrical Conductivity (EC)	EPA 120.1 and	1.95	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	35.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1250	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Live Oak Dairy C
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 1	T3240022B	Electrical Conductivity (EC)	EPA 120.1 and	1.68	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	30.2	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	960	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	33.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1070	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)			3.04
Electrical Conductivity (EC)	93.9			1.46 (T3240023E)	1.46 (Dup)	0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)			2.36
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Live Oak Dairy C
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 9.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 2	T3240022C	Electrical Conductivity (EC)	EPA 120.1 and	1.73	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.7	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	885	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	12.7	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)			3.04
Electrical Conductivity (EC)	93.9			1.46 (T3240023E)	1.46 (Dup)	0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)			2.36
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 3	T3240022D	Electrical Conductivity (EC)	EPA 120.1 and	1.10	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.12	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	560	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	7.12	mg/L
		Soluble Salts (SALT-SOL)	Calculation	705	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)			3.04
Electrical Conductivity (EC)	93.9			1.46 (T3240023E)	1.46 (Dup)	0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)			2.36
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW-River	T3240022E	Electrical Conductivity (EC)	EPA 120.1 and	2.43	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.87	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1460	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	5.87	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1550	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)			3.04
Electrical Conductivity (EC)	93.9			1.46 (T3240023E)	1.46 (Dup)	0.343
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)			2.54
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:31 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 4	T3240022F	Electrical Conductivity (EC)	EPA 120.1 and	1.09	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.07	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	570	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	5.07	mg/L
		Soluble Salts (SALT-SOL)	Calculation	695	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)			3.04
Electrical Conductivity (EC)	93.9			1.46 (T3240023E)	1.46 (Dup)	0.343
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)			2.54
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live oak C
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: B.N
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	8/25	8:40	Domestic well	DW#1	10.6'		✓				
2	8/25	8:15	Irrigation wells	IW 1	11.8'						H7
3	8/25	8:25		IW 2	9.8'						H7
4	8/25	8:35		IW 3	11.0'						H7
5	8/25	8:45		IW-RIVER		12.0'					H7
6	8/25	8:50		IW 4		11.0'					H7
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER (LABORATORY USE ONLY)
1st						
2nd		DENELE ANALYTICAL, INC.	8-28-23 @ 12:31			T3240022

Required to be submitted to any agency? (YES / NO)

2023 Annual Report LOC

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAniurorO5B8chZJkDJt2dNUlnSqy3AayN

"2023 Annual Report LOC" History

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